Freight Rate Prediction

Technical assessment report | Prepared for the Machine Learning Engineer evaluation | 27 October 2026

48,000	12,000	$115.66
Labeled development rows	Final rate predictions	October holdout MAE

Executive summary

The provided January–October 2025 labeled freight records were used to build and evaluate rate estimators. Four LightGBM candidates were selected using September data only. The best model was assessed once on a separate October holdout, then refitted on all 48,000 labeled loads to predict 12,000 November–December loads.

The winning model is reduced + log: a reduced-input LightGBM regressor trained on log(1 + posted_rate), using 117 boosting trees. It achieved $115.66 MAE and 4.86% WAPE in the October holdout. Its MAE was 54.4% lower than a historical median rate-per-mile baseline.

What these results do and do not establish

All validation metrics in this report were computed from labeled development records. The 12,000 final validation rows have no posted_rate labels in the supplied files; therefore their prediction accuracy cannot be measured here. The supplied score.py checks IDs, output format and chart inputs, not hidden accuracy.

Delivered artifacts

Artifact	What it contains
validation_predictions.csv	All 12,000 load IDs and predicted rates
december_predictions.csv	31 fixed-lane December daily estimates
scorer_results/candidate_december.png	December line chart generated by the supplied scorer
model_pipeline.py	Training, model selection, holdout validation and predictions
README.md / requirements.txt	Setup and reproducibility instructions

01 | Dataset and data-quality review

Training covers 2025-01-01 through 2025-10-31. The prediction file spans 2025-11-01 through 2025-12-31; the date ranges do not overlap. Source fields include shipment locations, distance, equipment, weight, date and supplementary market indicators. Only training contains the target posted_rate.

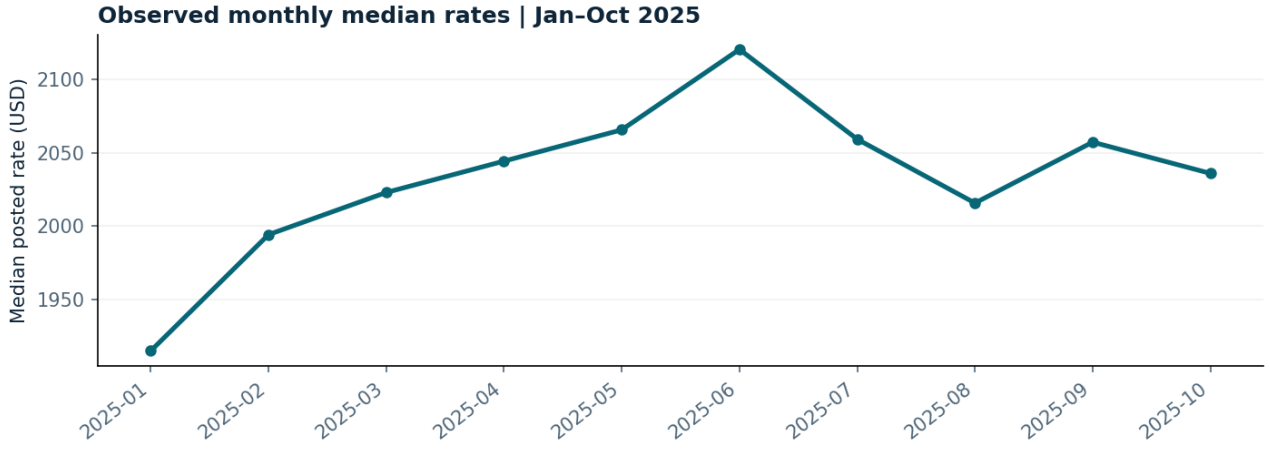
Observed issue	Train data	Final prediction data
Missing weight	300	165
Weight at or below zero	292	145
Missing market_index	374	249
Cities unseen in the training set	—	8 distinct city names
Posted rates above $10,000	103	Not labeled

Handling decisions

Weights at or below zero were treated as missing rather than replaced with invented values. LightGBM handles missing numeric features. Missing city labels are represented as an unknown category; cities not seen in the fitted data are treated as missing categorical values. The target contains no invalid/nonpositive values.

Training posted rates range from $57.22 to $25,533.00. Large observations were retained: they are present in the supplied labeled file, and there is no evidence in that file that they are errors.

Observed data pattern



The plot shows the observed monthly median of posted_rate across the original training records, not model output. Its variation can reflect both changes in load mix and changes in rates.

02 | Time-aware model validation

A chronological split was used to represent deployment on future loads and avoid the future-to-past leakage that can occur with random splitting. Feature engineering is deterministic and row-wise; posted_rate never appears among model inputs.

Period	Rows	Use
Jan–Aug 2025	38,477	Fit model candidates
September 2025	4,670	Early stopping and model selection
October 2025	4,853	Independent final offline holdout
January–October 2025	48,000	Final refit for submission

September candidate comparison

LightGBM variant	Trees	MAE	RMSE	WAPE
reduced / raw	91	$134.79	$617.65	5.60%
reduced / log	**117**	**$117.84**	**$618.77**	**4.90%**
full / raw	217	$142.46	$624.99	5.92%
full / log	237	$128.14	$624.08	5.32%

Model selection criterion: lowest September dollar MAE; boosting rounds chosen by early stopping on September. Reduced variants exclude coordinates, market_index and quote_signal. Full variants use those fields as well. The selected reduced/log model therefore needs no unavailable market or location fields for the December scenario.

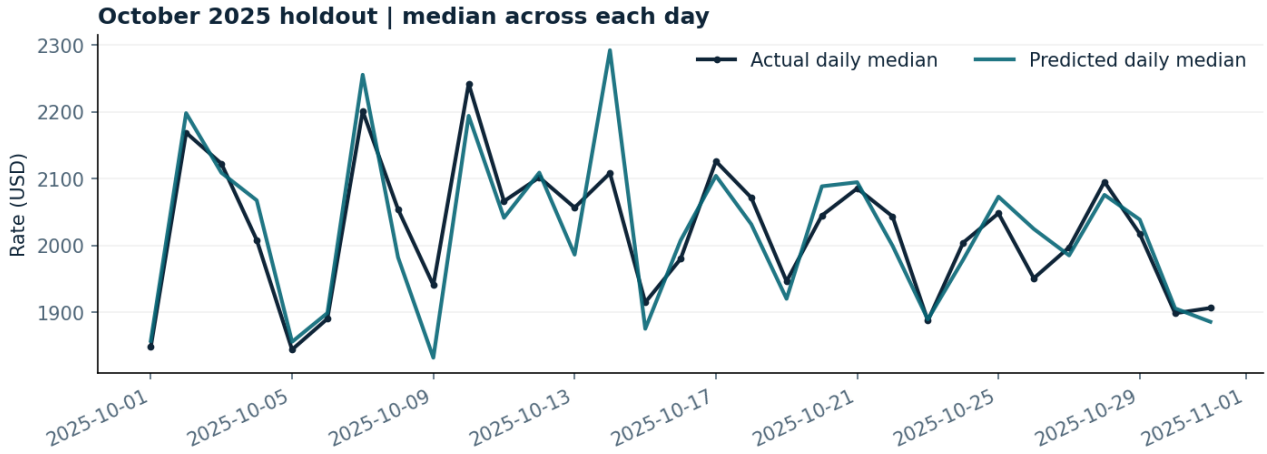
October holdout | selected model vs simple baseline

Method	MAE	RMSE	R^2	WAPE
Median historical rate per mile	$253.85	$695.05	0.793	10.67%
Selected LightGBM	**$115.66**	**$650.54**	**0.819**	**4.86%**

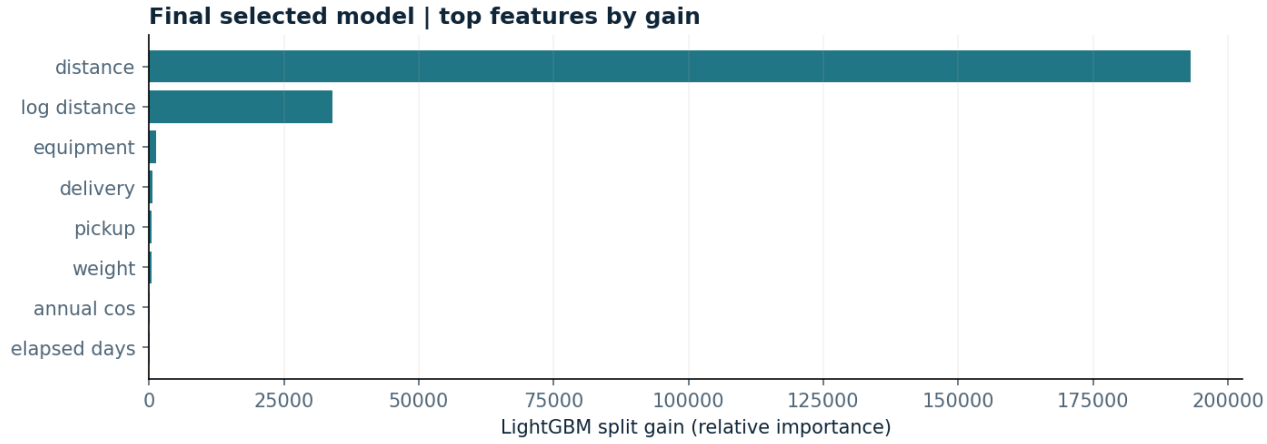
MAE measures mean absolute error in USD; RMSE penalizes large misses more heavily; R^2 measures the explained share of holdout variance; WAPE = sum(abs(error)) / sum(abs(actual)) × 100. October was not used to pick the model or tree count.

03 | Model diagnostics and interpretation

The selected regressor is trained on log(1 + rate), with predicted values converted back to USD using expm1. The model uses pickup, delivery, equipment, distance, weight, calendar variables and derived numeric transformations. It was retrained on all labeled January–October loads only after October evaluation.



Each point summarizes the median of the actual or predicted October loads for that day. This diagnostic is descriptive and is not an independent accuracy metric beyond the October holdout scores.



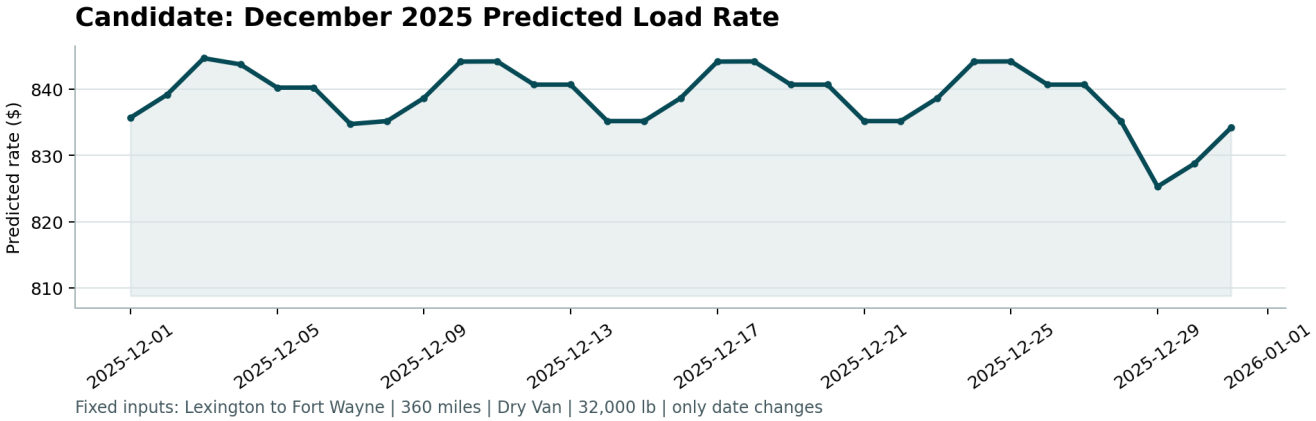
Gain is model-internal LightGBM feature importance, not a causal effect or evidence of an independently verified price driver. Long-haul distance dominates the fitted model.

Key limitations

Holdout RMSE ($650.54) substantially exceeds MAE ($115.66), indicating some relatively large forecast errors. Eight city names appear in the final prediction data but not training. Forecasting beyond October can also encounter shifts in holiday patterns, routing mix or market conditions.

04 | Fixed December 2025 prediction chart

The assessment specifies the same load on each day of December 2025: Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date changes. The reported values were generated by the selected final reduced-feature model and the graph below was rendered by the provided score.py without changing its fixed configuration.



The daily predictions range from $825.30 to $844.68. These are estimates from a time-dependent model, not observed December prices. No external quotes or future market measurements were supplied or invented.

Reproduction instructions

Step	Action
1	Install dependencies: python -m pip install -r requirements.txt
2	Place the four original assessment CSVs in the data/ directory
3	Run python model_pipeline.py to reproduce predictions and evaluation
4	Run python score.py --predictions validation_predictions.csv --december-predictions december_predictions.csv

Scorer outcome: 12,000 final prediction rows accepted; all 31 December records accepted; candidate_december.png created. The submission additionally requires an accessible GitHub repository and a separate 2–3 minute camera-on Loom video.

Source: employer-provided CSVs and score.py. All statistics, evaluation scores and plotted model outputs were generated from these files by the included Python pipeline.